

Query Explanations

Question 1

This query counts the total number of GradCafe entries for Fall 2026 using the term field. It provides an estimate of the size of the Fall 2026 applicant dataset and serves as a baseline for several later calculations.

Question 2

This query calculates the percentage of applicants identified as international students. The number of international applicants is divided by the total number of applicants and multiplied by 100 to obtain a percentage.

Question 3

This query calculates the average GPA, GRE Total, GRE Verbal, and GRE Analytical Writing scores. Invalid placeholder values and missing values are excluded to ensure the averages better reflect actual applicant data.

Question 4

This query calculates the average GPA among American applicants who applied for Fall 2026. The query filters applicants by both nationality and application term before calculating the average GPA.

Question 5

This query calculates the acceptance rate for Fall 2026 applicants. The number of accepted applicants is divided by the total number of Fall 2026 applicants and reported as a percentage.

Question 6

This query calculates the average GPA of applicants who were accepted for Fall 2026 admission. It allows comparison between accepted applicants and the overall applicant pool.

Question 7

This query counts the number of applicants who applied to Johns Hopkins University for a Master's degree in Computer Science. The query filters records using university name, degree type, and program name.

Question 8

This query counts accepted Fall 2026 PhD Computer Science applicants who applied to Georgetown University, MIT, Stanford University, or Carnegie Mellon University. Multiple university filters are combined in a single SQL query.

Question 9

This query repeats Question 8 using LLM-generated standardized fields rather than the original text fields. The purpose is to verify that the data cleaning and standardization process produces the same analytical result.

Question 10

This query counts applicants who applied to Johns Hopkins University for biology-related PhD programs. Program names are filtered using standardized program classifications generated during data cleaning.

Question 11

This query calculates the acceptance rate among Johns Hopkins University biology-related PhD applicants. The number of accepted applicants is divided by the total number of biology-related PhD applicants and reported as a percentage.

Additional Notes

All queries were executed directly against the PostgreSQL applicants table. SQL aggregation functions such as COUNT(), AVG(), and ROUND() were used extensively. Filtering was performed using WHERE clauses based on application term, status, university, degree, nationality, and program fields. The Flask dashboard displays the results dynamically and allows the analysis to be refreshed after new data is added to the database.